

ProctorShield AI — System Design Document (SDD)

Software Architecture, Data Design, Real-Time Subsystems & Security Specifications

ProctorShield AI — Official System Documentation | Generated: August 2026

1. Executive Summary & Technology Stack

ProctorShield AI is a fullstack web platform for remote exam proctoring and integrity monitoring. Built on Next.js 16 (App Router), React 19, TypeScript, and TailwindCSS v4, the system combines real-time WebRTC P2P video streaming, client-side computer vision ML (@vladmandic/face-api, COCO-SSD), server-side Gemini 2.5 Flash AI, and Neon PostgreSQL via Prisma ORM.

2. System Architecture & Layer Diagram

The system follows a decoupled 5-tier architecture:

- Client Interface Layer: Next.js 16 App Router UI with Figma-aligned dark proctoring workspace, 3D Head Direction Radar compass, and responsive dashboards.
- WebRTC P2P & WebSocket Signaling Subsystem: Direct student-to-teacher 60 FPS HD video streaming backed by Pusher channels for signaling and notifications.
- Client-Side AI Computer Vision Engine: @vladmandic/face-api 68-landmark 3D pitch/yaw head vector calculation, COCO-SSD cellphone detection, and Web Audio API RMS analyzer.
- Server API & AI Verdict Subsystem: Next.js Route Handlers processing real-time violation logs, session authentication, email dispatches, and Gemini 2.5 Flash verdict generation.
- Persistence Layer: Prisma ORM connected to Neon PostgreSQL hosted cloud database.

3. Real-Time Video Subsystem Specifications

The video monitoring system features a Dual-Engine Architecture:

- Engine A (WebRTC P2P 60 FPS HD): Direct RTCPeerConnection stream offering ultra-smooth 60 FPS video with zero server relay costs.
- Engine B (1.5s High-Frequency Snapshot Stream): Automatic HTTP snapshot fallback uploading camera frames every 1.5 seconds to /api/live/snapshot. Resolves single-browser multi-tab session testing and restricted firewalls.
- 5-Second WebM Video Evidence Recording: MediaRecorder API capturing 5-second video clips upon violation detection, viewable in HTML5 Evidence Replay.

4. Database Schema Specifications (Prisma ORM)

Model Name	Primary Key	Description	Key Relations
User	id (UUID)	User accounts (Student/Teacher...	Role, StudentQuiz, Quiz
Role	id (Int)	RBAC Roles (admin, teacher, st...	User[]
Subject	id (Int)	Academic Subject Course	User (Teacher), Quiz[]
Quiz	id (Int)	Exam Configuration & Timer	Subject, User, Question[]
Question	id (Int)	Quiz Questions & Points	Quiz, Choice[], Answer[]
Choice	id (Int)	Multiple Choice Answers	Question
StudentQuiz	id (UUID)	Student Exam Session & Score	User, Quiz, Violation[]
Violation	id (BigInt)	Cheating Incident & Video Evid...	StudentQuiz, EvidenceFile[]
EvidenceFile	id (BigInt)	Uploaded Media Evidence Path	Violation
AiAnalysis	id (Int)	Gemini AI Verdict & Probability	StudentQuiz

Model Name	Primary Key	Description	Key Relations
Notification	id (BigInt)	User Alerts & Notifications	User
ActivityLog	id (BigInt)	Audit Trail & User Actions	User

5. Security, Lockdown & Control Flow

Anti-Cheating Lockdown System:

- Fullscreen Lockdown: Automatic `document.documentElement.requestFullscreen()` on quiz start.
- Startup Grace Period: 5-second initial grace period (`isStartupGracePeriodRef`) preventing false `window_resize` violations during startup.
- Pre-Warning System: 2.5s–3s warning banner delay before logging official violation strikes for tab switching or fullscreen exit.
- 3-Strike Auto-Termination: Quiz auto-submits upon 3 strikes or severe cheating detection.
- Database Storage Management: DELETE `/api/dashboard/teacher/evidence` route allowing 1-click purging of violation evidence from Neon DB.